

1. CURRENT PROGRESS (COMPLETED)

- [+] Match results + goals synced from Google Sheets
 - Full-time scores (FTHG/FTAG) pulled from nwfl-regular sheet
 - Goal scorers with minute, penalty, and own-goal flags
 - Notes column synced for match postponements and admin remarks
- [+] Season-based navigation
 - Season field on every match (2021/22, 2023/24, 2024/25, 2025/26)
 - API filters matches by ?season= parameter
 - /api/matches/seasons/ endpoint returns unique season list
 - Admin season selector defaults to latest season
- [+] Multi-season data support
 - Backfilled 460+ matches across 4 seasons
 - Sync script reads Season column per row from Google Sheets
 - No more hardcoded season defaults
- [+] AI parsers for WhatsApp text, images, and bulk match entry
 - Single-match AI parsing from raw text
 - Bulk match parsing from text or image upload
 - Vision-based scoreboard extraction (when available)
- [+] Basic analytics endpoints
 - Goal distribution charts
 - Game state analytics
- [+] Admin platform
 - Full invitation system with role-specific emails
 - Auto-login after invite acceptance
 - Match CRUD with season, notes, and goal entry
 - Sheet sync with background processing and status logs
 - AI Import panel for WhatsApp/text/image match entry
- [+] Performance fixes
 - MatchListSerializer (lightweight, no nested goals) prevents timeouts
 - Pagination disabled on matches for full season views

2. CURRENT DATA GAPS

Data Type	Status	Notes
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Standings Auto-Calculation Missing Standings table exists but is NOT auto-recalculated from match results. Must be manually triggered or recalculated via management command.

Player Profiles Missing Goal scorers stored only as strings. No player stats, bios, positions, or persistent player records.

Live Match Events Missing No substitutions, cards, halftime scores, possession, shots, or in-game tracking. HTHG/HTAG columns exist in sheet but are not yet imported.

Fixtures / Schedule Partial Season filtering separates seasons, but upcoming matches are mixed with completed ones. No dedicated "fixtures-only" view.

Venue Data Missing Venue is only a text string. No stadium metadata, capacity, location, or maps.

Referee Data Missing Officials are not tracked.

Transfer / Loan Data Missing No player movement tracking.

Injury / Suspension Missing No player availability tracking.

Scrapers / External APIs Missing No automated ingestion from NWFL site, sports blogs, X, YouTube, or Facebook.

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3. AI INFRASTRUCTURE REALITY CHECK

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[+] Validation layer (PARTIAL)

- Deduplication checks for invitations
- Team name fuzzy-matching during sheet sync
- No validation layer for AI outputs before DB write (still needed)

[-] No web crawling

Current AI only parses text manually provided by users.
Cannot browse X, Facebook, YouTube, or Google in real time.

[-] Free model limitations

Current free-tier models have rate limits, no SLA guarantees, inconsistent output formats, and unreliable structured output.

- ☐ No retry/fallback logic
Single API failures can break sync operations.

4. PRODUCTION-GRADE ROADMAP

Phase 1 -- Foundation Stabilization (2-3 Weeks) [IN PROGRESS]

- ☐ Auto-recalculate standings from match results after every sync
- ☐ Add fixtures endpoint separating upcoming vs completed matches
- ☐ Create proper player model with persistent records and statistics
- ☐ Create match events model (substitutions, cards, halftime data)
- ☐ Import HTHG/HTAG (half-time scores) from sheet
- ☐ Add validation layer before AI outputs are written to database
- ☐ Build venue model with stadium metadata and location

Phase 2 -- Data Ingestion Expansion (1-2 Months)

- ☐ Build scrapers for NWFL official site, sports blogs, and media outlets
- ☐ Integrate X (Twitter) API for live club updates
- ☐ Integrate YouTube Data API for highlights and goal compilations
- ☐ Integrate Facebook Graph API for club page updates
- ☐ Aggregate RSS feeds from sports publishers
- ☐ Add computer vision pipeline for extracting scoreboards from images/videos

Phase 3 -- Monetization & Platform Expansion (2-3 Months)

- ☐ Launch public API as a paid data product
- ☐ Build embeddable widgets for blogs and news sites
- ☐ Offer premium analytics, predictions, and fantasy integrations
- ☐ Develop white-label admin infrastructure for other leagues

5. STRATEGIC DIRECTION

The immediate priority is stabilizing and structuring the data layer before expanding AI ingestion capabilities or monetization features.

Without strong validation, proper schemas, and automated recalculation systems, scaling data ingestion will increase inconsistency and technical debt.

Focus areas:

1. Reliable structured data models (players, venues, events)
2. Automated standings and fixture systems
3. Player and event-level tracking

4. Validation and sync reliability

Once the foundation is stable, the platform can evolve into a scalable football data infrastructure product for the NWFL ecosystem.

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